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Economics A: Microeconomics I, Group 2

Fall Semester 2026

Part 1.2

Gains from Trade

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Microeconomics I: Where We Are

Part 1: Introduction

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Part 1.2: Gains from Trade

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Part 3: Firm Behavior and Industrial Organization

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Part 1.2

Gains from
Trade



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Learning Goals

This chapter covers ...

- ▶ the application of the concept of **opportunity costs** toward understanding why the process of specialization and trade is potentially beneficial.
- ▶ why the principle of **comparative advantage** is crucial to an understanding of why societies organize economic activities and to the economic role of institutions.
- ▶ why **institutions** matter and what an economic theory of institutions has to explain in order to provide a better understanding of societal phenomena like growth, unemployment, globalization, or anthropogenic climate change.

Literature

- ▶ Kolmar (2022), *Principles of Microeconomics*, Ch. 2.
- ▶ Kolmar/Hoffmann (2022), *Workbook for Principles of Microeconomics*, Ch. 2.

The Theory of
Comparative
Advantage

Introductory
Examples

Three Questions
about Trade

Recap

Summary

Your Turn



- ▶ Fundamental aspect of coping with scarcity: the acts of individuals are **interdependent**.
- ▶ Basis for the phenomena of **exchange and specialization** (trade).
- ▶ Situation without trade: **autarky**; each individual produces everything he or she consumes.



The Theory of Comparative Advantage

Microeconomics I

Part 1.2

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Adam Smith (1723–1790)

division of labor, absolute advantage

(*Wealth of Nations*, 1776)



David Ricardo (1772–1823)

comparative advantage

(*Principles of Political Economy and Taxation*, 1817)

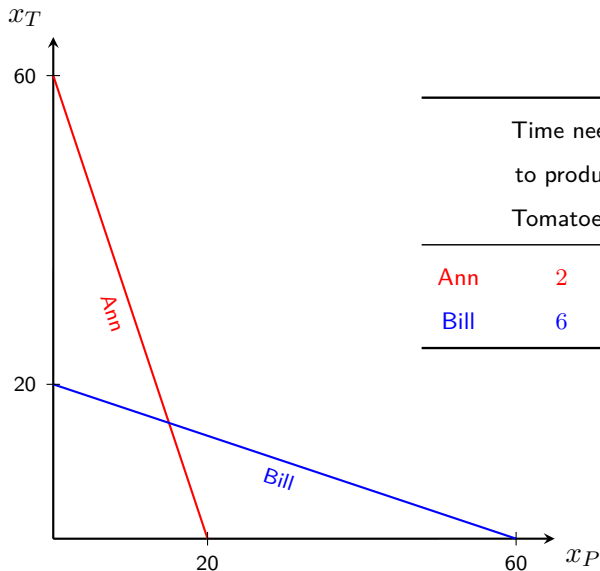


Two individuals (Ann and Bill), two goods (tomatoes and pears).

- ▶ Case 1: Each individual can produce exactly one good and wants to consume only that good.
- ▶ Case 2: Each individual can produce exactly one good and wants to consume both.
- ▶ Case 3: Each individual can produce both goods and wants to consume both. However, one individual can produce more tomatoes and the other can produce more pears.



Case 3: Production Possibilities for Ann and Bill



	Time needed (in h) to produce 1 kg of		Maximum production (in kg) of	
	Tomatoes	Pears	Tomatoes	Pears
Ann	2	6	60	20
Bill	6	2	20	60

Time budget: 120 h each.



Case 3: The Production-Possibility Frontier (PPF)

Definition: Production-Possibility Frontier

The PPF is the set of all combinations (x_P, x_T) of pears and tomatoes that an individual can produce when the time endowment (here: 120 h) is fully used.

Ann

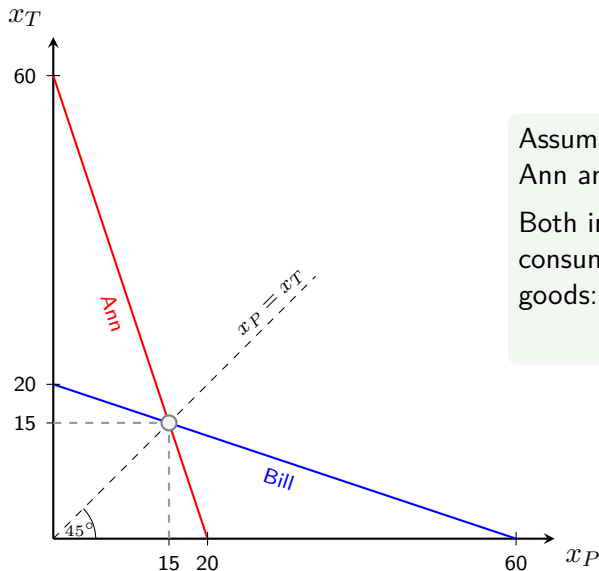
$$\text{Time budget: } 2 \cdot x_T + 6 \cdot x_P = 120 \quad \Leftrightarrow \quad x_T = 60 - 3 \cdot x_P$$

Bill

$$\text{Time budget: } 6 \cdot x_T + 2 \cdot x_P = 120 \quad \Leftrightarrow \quad x_T = 20 - \frac{1}{3} \cdot x_P$$



Case 3: Production and Consumption in Autarky (1/2)



Assumption regarding the preferences of Ann and Bill:

Both individuals always want to consume the same amount of both goods:

$$x_{\text{Tomatoes}} = x_{\text{Pears}}.$$

Case 3: Production and Consumption in Autarky (2/2)

In autarky, consumption equals production: the preference $x_T = x_P$ must hold on each individual's PPF.

Ann

PPF and preference: $x_T = 60 - 3 \cdot x_P$ and $x_T = x_P$

$$\Rightarrow x_P = 60 - 3 \cdot x_P \Leftrightarrow x_P = x_T = 15$$

Bill

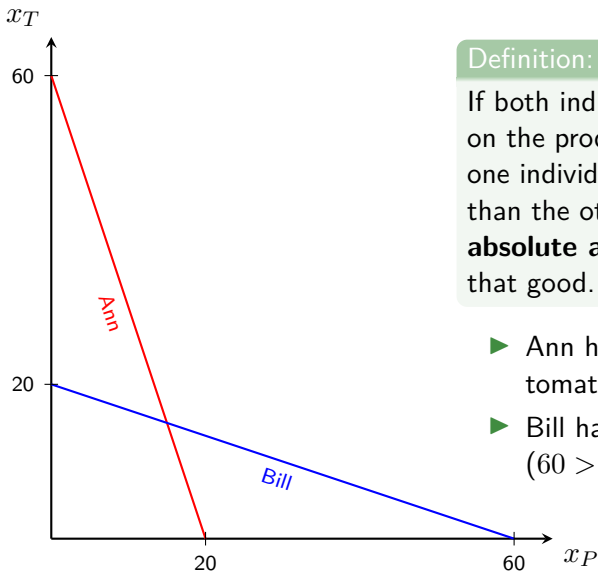
PPF and preference: $x_T = 20 - \frac{1}{3} \cdot x_P$ and $x_T = x_P$

$$\Rightarrow x_P = 20 - \frac{1}{3} \cdot x_P \Leftrightarrow x_P = x_T = 15$$

In this (symmetric) example, both produce and consume (15, 15) in autarky: the intersection of the 45° line with the respective PPF.



Case 3: Absolute Advantage



Definition: Absolute Advantage

If both individuals concentrate exclusively on the production of the same good, and one individual produces more of that good than the other, then this individual has an **absolute advantage** in the production of that good.

- ▶ Ann has an absolute advantage in tomatoes ($60 > 20$).
- ▶ Bill has an absolute advantage in pears ($60 > 20$).

Case 3: Specialization Along Absolute Advantage

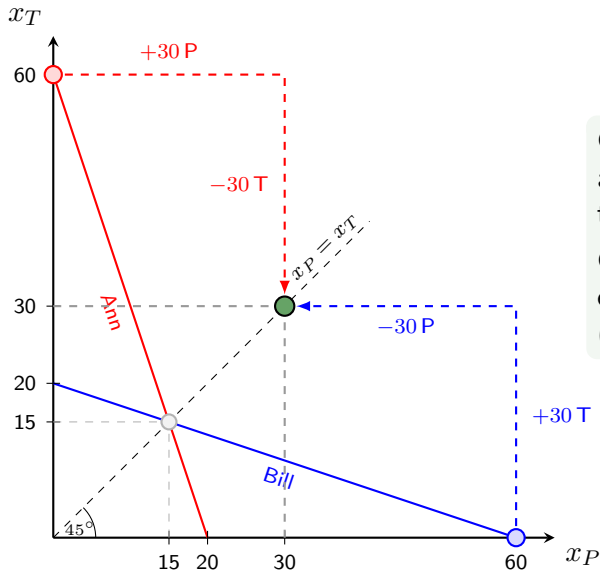
Suppose Ann and Bill change their production *in the direction of their absolute advantage*: Ann produces only tomatoes, Bill produces only pears.

Production	Tomatoes (in kg)	Pears (in kg)
Autarky: Ann / Bill	15 / 15	15 / 15
Σ Autarky	30	30
Specialization: Ann / Bill	60 / 0	0 / 60
Σ Specialization	60	60

Overall output **doubles**, from (30, 30) to (60, 60), although nobody works more.



Case 3: Gains from Trade



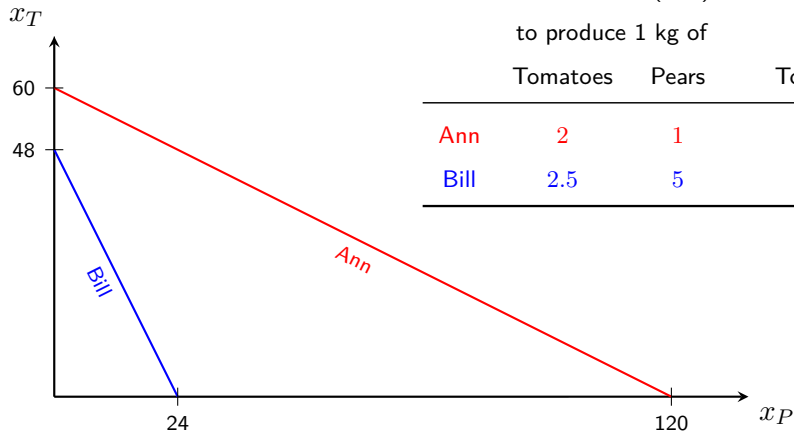
Given the new production allocation, Ann trades 30 kg of tomatoes for 30 kg of Bill's pears.

Given the preferences $x_T = x_P$, each consumes (30, 30) instead of (15, 15): **both are better off.**

- ▶ Case 4: Each individual can produce both goods and wants to consume both. However, one individual can produce both more tomatoes and more pears.
- ▶ Ann now has an absolute advantage in *both* goods.
- ▶ Question: Can specialization and trade still be mutually beneficial?



Case 4: Production Possibilities for Ann and Bill



	Time needed (in h) to produce 1 kg of		Maximum production (in kg) of	
	Tomatoes	Pears	Tomatoes	Pears
Ann	2	1	60	120
Bill	2.5	5	48	24

Time budget: 120 h each.

Case 4: PPFs and Consumption in Autarky

The PPFs again follow from the time budgets; in autarky, consumption equals production, so the preference $x_T = x_P$ must hold on each PPF.

Ann

$$\text{Time budget: } 2 \cdot x_T + 1 \cdot x_P = 120 \quad \Leftrightarrow \quad x_T = 60 - \frac{1}{2} \cdot x_P$$

$$\text{Autarky } (x_T = x_P): \quad x_P = 60 - \frac{1}{2} \cdot x_P \quad \Leftrightarrow \quad x_P = x_T = 40$$

Bill

$$\text{Time budget: } 2.5 \cdot x_T + 5 \cdot x_P = 120 \quad \Leftrightarrow \quad x_T = 48 - 2 \cdot x_P$$

$$\text{Autarky } (x_T = x_P): \quad x_P = 48 - 2 \cdot x_P \quad \Leftrightarrow \quad x_P = x_T = 16$$

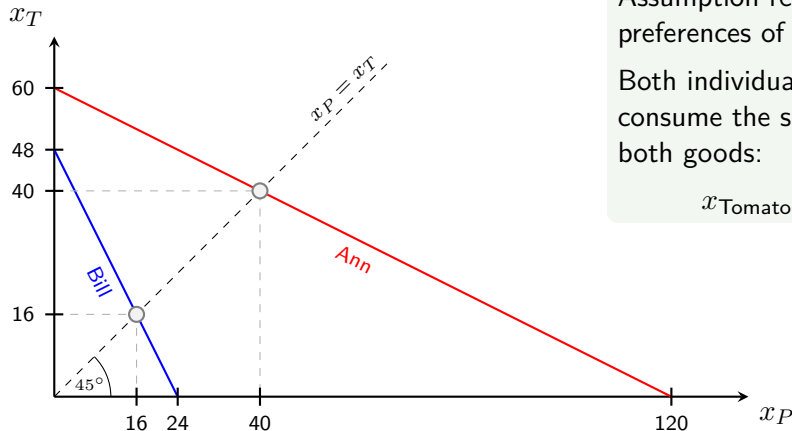


Case 4: Production and Consumption in Autarky



Assumption regarding the preferences of Ann and Bill:
Both individuals always want to consume the same amount of both goods:

$$x_{\text{Tomatoes}} = x_{\text{Pears}}.$$

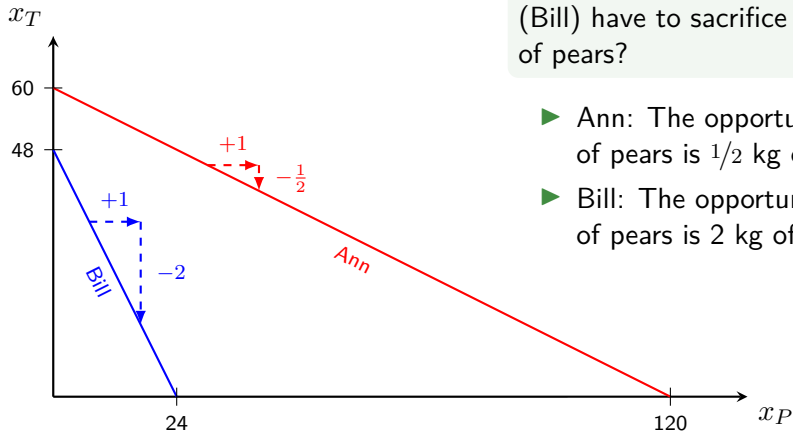


The Concept of Opportunity Costs (OC) (1/2)



How many kilos of tomatoes does Ann (Bill) have to sacrifice for another kilo of pears?

- Ann: The opportunity cost of 1 kg of pears is $\frac{1}{2}$ kg of tomatoes.
- Bill: The opportunity cost of 1 kg of pears is 2 kg of tomatoes.

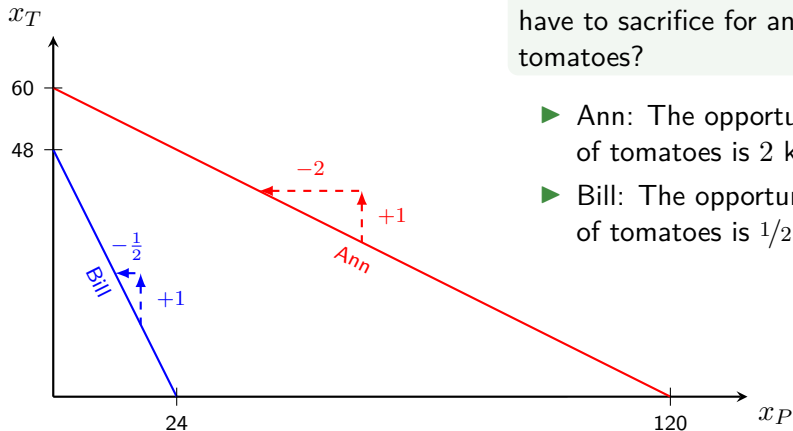


The Concept of Opportunity Costs (OC) (2/2)



How many kilos of pears does Ann (Bill) have to sacrifice for another kilo of tomatoes?

- Ann: The opportunity cost of 1 kg of tomatoes is 2 kg of pears.
- Bill: The opportunity cost of 1 kg of tomatoes is $\frac{1}{2}$ kg of pears.



Comparative Advantage

Definition: Comparative Advantage

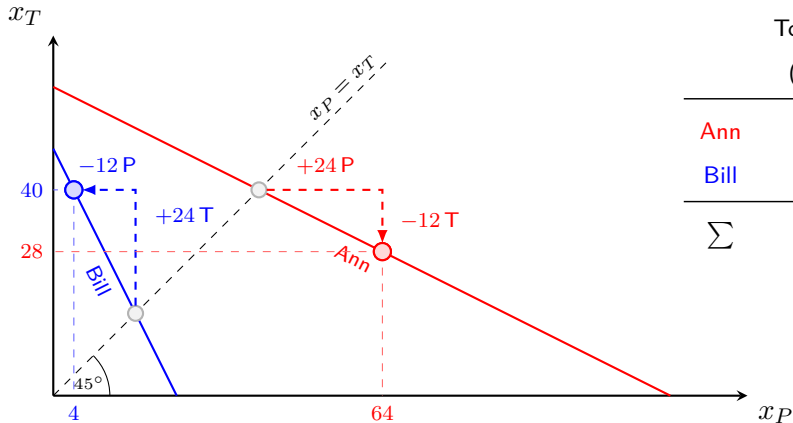
An individual has a **comparative advantage** in the production of a good if he or she can produce it at lower opportunity costs, measured in units of the other good, than the other individual.

	Opportunity costs for 1 kg	
	Tomatoes	Pears
	(in kg Pears)	(in kg Tomatoes)
Ann	2	$\frac{1}{2}$
Bill	$\frac{1}{2}$	2

- ▶ Ann's opportunity costs of 1 kg of pears ($\frac{1}{2}$ kg of tomatoes) are lower than Bill's (2 kg): **comparative advantage in pears**.
- ▶ Bill's opportunity costs of 1 kg of tomatoes ($\frac{1}{2}$ kg of pears) are lower than Ann's (2 kg): **comparative advantage in tomatoes**.

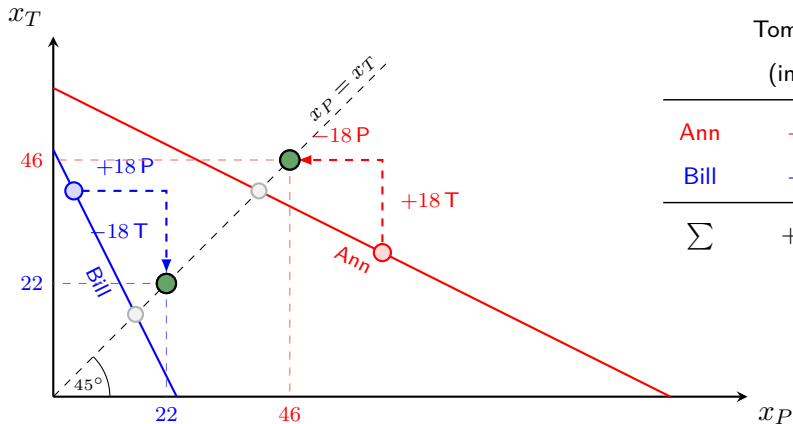


Specialization and Trade I: Change in Production



	Change in production	
	Tomatoes (in kg)	Pears (in kg)
Ann	-12	+24
Bill	+24	-12
Σ	+12	+12

Specialization and Trade II: Trade and Change in Consumption



	Change in consumption	
	Tomatoes (in kg)	Pears (in kg)
Ann	+6	+6
Bill	+6	+6
Σ	+12	+12

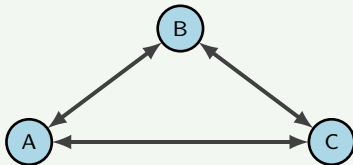
Three Questions about Trade

1. Does the process of specialization and trade make everybody better off?
2. Will gains from trade be exploited automatically, or do individuals have to create specific “environments” (institutions)?
3. Is the example robust in the sense that it represents an important aspect of reality, or is it valid only under specific conditions?

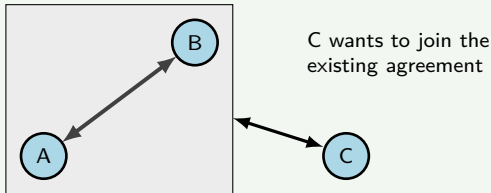


1st Question: Simultaneous vs. Sequential Integration

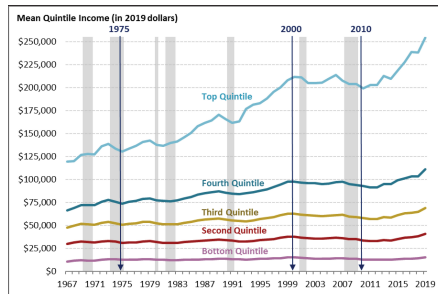
Simultaneous Integration



Sequential Integration



1st Question: The US Income Distribution



Mean Quintile Household Income, 1967-2019

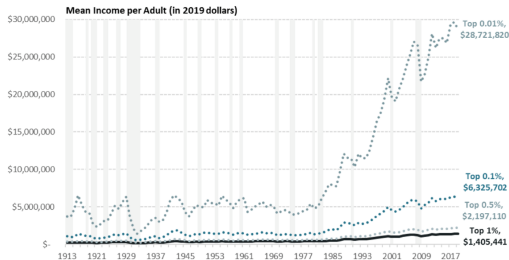
Source: [Donovan et al. \(2021\)](#), *The U.S. Income Distribution: Trends and Issues*,

Congressional Research Service, R44705.

- ▶ USA: From the mid-1970s to 2000, income inequality increased significantly, yet incomes grew in *all* quintiles.
- ▶ Between 2000 and 2010, average real household income *declined in all quintiles*.
- ▶ Between 2010 and 2019, it recovered in each quintile, while inequality resumed its rise.



1st Question: Top Incomes and Their Drivers



Mean Income per Adult, Select Percentiles, 1913-2019

Source: [Donovan et al. \(2021\)](#), *The U.S. Income Distribution: Trends and Issues*,

Congressional Research Service, R44705.

- ▶ Since the late 1970s, top incomes have decoupled from the rest of the distribution.
- ▶ The median and lower percentiles grew much more slowly over the same period.
- ▶ The gains from integration were thus real, but very unevenly distributed.



2nd Question: Trade and Conflict

Two competing theories regarding specialization, trade, exploitation and conflict:

Theory 1

Specialization and trade **make international (military) conflicts less likely** as countries increasingly depend on each other (globalization as a peacemaker).

Theory 2

Specialization and trade **lead to dependencies that can be uni- or multilaterally exploited**. Interdependencies can make even military conflicts more likely for mainly two reasons:

- ▶ Economic dependencies can be asymmetrically exploited.
- ▶ Some countries specialize in indispensable resources or intermediate products.



2nd Question: The Hold-Up Problem

The Hold-Up Problem

*“Economists have coined the term **hold-up problem** for a situation where specialization creates power asymmetries that can be exploited by the relatively more powerful partner.”*

Kolmar (2022), *Principles of Microeconomics*, p. 40

Whether **pacification** (Theory 1) or **exploitability** (Theory 2) dominates is, ultimately, an *empirical* question.



2nd Question: Dependencies Today



Natural gas pipelines from Russia to Europe

Russia: ~40 % of EU gas imports (2021)
→ ~8 % (2023)

Source: [Wikipedia](#),
[List of natural gas pipelines](#)

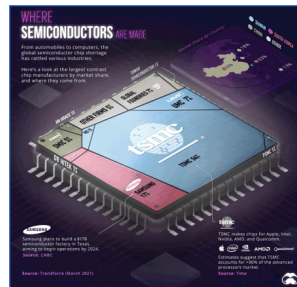


*China Has Paid a High Price for Its
Dominance in Rare Earths*

Dust and groundwater contaminated with heavy metals and radioactive chemicals pose a health threat that the authorities have been trying to address for years.

Rare Earths

Source: [Keith Bradsher \(2025\) NYT](#)



Global dependency on Taiwanese
semiconductors (TSMC ~54 % foundry
share)

Source: [TrendForce \(2021\)](#) /
[Visual Capitalist](#)

- ▶ The institutions that structure the exchange are important.
- ▶ Specialization leads to dependencies that can be exploited.
- ▶ This is a problem for countries as well as for firms and individuals.



2nd Question: Why Gains from Trade Need Institutions

- ▶ Gains from trade do not materialize automatically: specialization creates interdependencies, and thus vulnerability (**hold-up**).
- ▶ Whether trade makes everybody better off depends on the *rules* under which it takes place.

*“Economics, as the study of how men and society choose to employ scarce productive resources, is therefore primarily **the study of institutions**.”*

Kolmar (2022), *Principles of Microeconomics*, p. 44



3rd Question: Dynamic Comparative Advantage

Theory of Dynamic Comparative Advantage

- ▶ The classical theory of comparative advantage (Ricardo) is *static*: it takes current resources and technologies as given.
- ▶ However, investments, learning, and government policy can change a country's comparative advantage over time.

Friedrich List and the *Infant-Industry* Argument

- ▶ Core idea: comparative advantage is not fixed; it can be altered through political measures.
- ▶ He criticized the free trade doctrines of Adam Smith and David Ricardo.



FRIEDRICH LIST
1789–1846



3rd Question: List's Infant-Industry Program

Friedrich List and the *Infant-Industry* Argument

- ▶ He proposed a *National System of Political Economy* (1841):
 - ▶ Protection of *young industries* through tariffs or subsidies.
 - ▶ Allowing these industries to grow until they are globally competitive.
 - ▶ Then reducing protection to enable competition.
- ▶ Logic: Short-term losses (higher prices, lower efficiency) can lead to long-term gains (higher productivity, new comparative advantage).
- ▶ Relevance today: Protectionism is often defended as a means of industrialization or re-industrialization; whether it *works* is intensely debated (market vs. government failures; cf. Digression 2.2).



3rd Question: Back to Bill

Model: Back to Bill, who only has a basic education

- ▶ With only a basic education, Bill's *opportunity costs* are lowest in simple tasks (delivery, cleaning): his comparative advantage lies there, even though these jobs pay little (low *absolute* productivity).
- ▶ Ricardo's logic: specialize where your *opportunity cost* is lowest today.
- ▶ But:
 - ▶ Bill invests in education/training.
 - ▶ Over time, Bill develops skills in programming, engineering, or medicine.
 - ▶ His comparative advantage shifts to high-productivity, well-paid work.



Is there reliable data for the *Infant Industry* argument? (1/2)

The Napoleonic Blockade

Juhász, Réka (2018)

Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade

American Economic Review 108(11): 3339–76



RÉKA JUHÁSZ

Temporary Protection & Technology Adoption

- ▶ Objective: identify the causal effect of temporary trade protection.
- ▶ Method: the Continental Blockade (1806–1814) as a natural experiment: Napoleon's *self*-blockade against British goods (Berlin Decree, 1806).
- ▶ British goods kept leaking in via smuggling hubs (Gibraltar, Malta): the **effective distance to London** rose sharply in the north, much less in the south: regionally different trade-cost shocks.



Is there reliable data for the *Infant Industry* argument? (2/2)

Conclusion

- ▶ High temporary protection led to significant growth and new establishments in the cotton industry of the northern regions.
- ▶ The south of France stagnated.
- ▶ **Persistence:** the northern advantage survived the *end* of protection: spinning activity remained concentrated in the north in 1840 and 1887.
- ▶ Caveat (Juhász): temporary protection *can* shift comparative advantage, not a general license for protectionism.



Recap: Answers to Our Three Questions

1. Does the process of specialization and trade make everybody better off?

Specialization and trade can create losers (cf. the US income-distribution data).

2. Are specific institutions needed?

Yes: specialization creates **hold-up** vulnerability; gains from trade depend on the rules of exchange: economics is “primarily the study of institutions” (Kolmar 2022).

3. Is the example robust?

The mechanism is general, but comparative advantage is *dynamic* and can be shifted (List, Juhász).



Summary: What You Should Take Away

- ▶ The acts of individuals are interdependent; exchange and specialization (trade) are a fundamental response to scarcity, autarky is the benchmark.
- ▶ The **PPF** shows all feasible production combinations for a given time endowment; its slope is the **opportunity cost**.
- ▶ **Absolute advantage**: producing more of a good than the other individual when both concentrate exclusively on it.
- ▶ **Comparative advantage**: lower opportunity costs; it, not absolute advantage, determines who should specialize in what (Ricardo).
- ▶ Even if one individual is better at everything, specialization along comparative advantage plus trade can make **both** better off.
- ▶ Gains from trade are not automatic: they require **institutions** (hold-up problem), and comparative advantage is **dynamic** (List, Juhász).



Original Exam Questions

Decide whether the following statements are *true* or *false*.



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#M1-P1-2-G2

